THE DEGREE IN WHICH A NORMALIZED DUALIZING COMPLEX IS NONZERO AT A MINIMAL PRIME

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The quotient by a power of a prime ideal of dimension zero inherits a normalized dualizing complex. Invoked inside the counted proof (source lines 3426). Replaces the ring by a quotient in which the ideal has become zero, without changing the localized complex.
- The degree confinement for the cohomology of a normalized dualizing complex. Invoked inside the counted proof (source lines 3428-3429). Shows that only the degree equal to minus the dimension can be nonzero.
- The localization of a dualizing complex at a prime is dualizing over the local ring. Invoked inside the counted proof (source lines 3430-3432). Shows the remaining cohomology module is nonzero.

Source reference. Original lemma lemma-dualizing-localize: sources/stacks-dualizing/pinned-original-source.tex, source lines 2950–2950. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing-localize, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-normalized-quotient: sources/stacks-dualizing/pinned-original-source.tex, source lines 3160–3160. Navigation pointer only: the counted text refers to the Stacks item lemma-normalized-quotient, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-sitting-in-degrees: sources/stacks-dualizing/pinned-original-source.tex, source lines 3232–3232. Navigation pointer only: the counted

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text refers to the Stacks item lemma-sitting-in-degrees, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-base-change-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2719–2719. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-base-change-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring with normalized dualizing complex $\omega_A^\bullet$. Let $\mathfrak{p}$ be a minimal prime of A with $\dim(A/\mathfrak{p}) = e$. Then $H^i(\omega_A^\bullet)_\mathfrak{p}$ is nonzero if and only if $i = -e$.*

Proof. Since $A_\mathfrak{p}$ has dimension zero, there exists an integer $n > 0$ such that $\mathfrak{p}^n A_\mathfrak{p}$ is zero. Set $B = A/\mathfrak{p}^n$ and $\omega_B^\bullet = R\operatorname{Hom}_A(B, \omega_A^\bullet)$. Since $B_\mathfrak{p} = A_\mathfrak{p}$ we see that

$$(\omega_B^\bullet)_\mathfrak{p} = R\operatorname{Hom}_A(B, \omega_A^\bullet) \otimes_A^{\mathbf{L}} A_\mathfrak{p} = R\operatorname{Hom}_{A_\mathfrak{p}}(B_\mathfrak{p}, (\omega_A^\bullet)_\mathfrak{p}) = (\omega_A^\bullet)_\mathfrak{p}$$

The second equality holds by More on Algebra, Lemma more-algebra-lemma-base-change-RHom. By Lemma lemma-normalized-quotient we may replace A by B. After doing so, we see that $\dim(A) = e$. Then we see that $H^i(\omega_A^\bullet)_\mathfrak{p}$ can only be nonzero if $i = -e$ by Lemma lemma-sitting-in-degrees parts (1) and (2). On the other hand, since $(\omega_A^\bullet)_\mathfrak{p}$ is a dualizing complex for the nonzero ring $A_\mathfrak{p}$ (Lemma lemma-dualizing-localize) we see that the remaining module has to be nonzero. □